

Math 170S - Homework 1 - due January 16

Review main concepts, definitions, and examples from 170E, including

- random variables, discrete and continuous distributions, mean and variance
- PMF (probability mass function = density function for discrete random variables), PDF (probability density function = density function for continuous random variables), CDF (cumulative distribution function: $f_X(t) = \mathbb{P}(X \leq t)$).
- Independence, mean and variance of sum or difference of independent random variables
- Conditional probability and Bayes theorem
- Examples of distributions, their interpretations, main properties: Uniform (discrete and continuous), Bernoulli, Binomial, Poisson, Geometric, Gaussian (= Normal), Exponential
- Properties of normal distribution
- Moment generating functions
- Central limit theorem

Disclaimer: This is a review homework, based on the material from probability class (only the last problem slightly touches the material of 6.1). You are encouraged to use the textbook Chapters 1-5, collaborate on all the problems (although the write-up of the solutions must be your own) and talk to me if you feel that some of the material is not familiar to you.

Solve the following problems. Unless specified otherwise, show all your work.

Problem 1. (Properties of standard distributions)

- Let X be a Binomial random variable such that $\mathbb{E}(X) = 4$ and $\text{Var}(X) = 2$. Find the parameters of X .
- Let X, Y be two independent standard normal random variables. What is mean and variance of $Z = 5X - 2Y$?
- Let X is $N(\mu, \sigma^2)$, what is the distribution of $Y = aX + b$? Here, a and b are some real numbers.
- Let $X_1, \dots, X_n$ be a independent normal variables with mean μ and variance $\sigma \geq 0$. Let us denote $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. What is the distribution of $\bar{X}$? Give the name of the distribution and find all its parameters.
- Let $X_1, \dots, X_n$ be a independent normal variables with different means μ_i and variances $\sigma_i \geq 0$. Let us denote $Z = \sum_{i=1}^n c_i X_i$, where c_i are some numbers. What is the distribution of Z ?

Note that part (d) is a particular case of part (e) (what are μ_i, σ_i and c_i , that make $\bar{X} = Z$?). Check that your answers for parts (d) and (e) well agree with each other.

Problem 2. (Bayes theorem) A store sells three types of shoes: running shoes (40% of all shoes in the store), hiking shoes (30%) and raining shoes (30%). It is known that 10% of running shoes are water proof; 70% of hiking shoes are water proof; all raining shoes are, of course, water proof. If a selected shoe turns out to be water proof, what is the probability that it is a hiking shoe?

Problem 3. (Central limit theorem) A food store just got a delivery of 10000 potatoes. It is known that each potato is rotten with probability 0.1. What is the expected number of healthy potatoes? Use the normal approximation to estimate the probability that there are at least 8970 healthy potatoes.

Problem 4. (Main properties of density functions) 3.1-20 from Hogg et al textbook.

Problem 5. (Chi-square distribution)

- (a) What is the density function, expectation and variance of the chi-square distribution with 4 degrees of freedom $\chi^2(4)$? (You can find the formulas in the back of your textbook. Look at page 99 if you are not clear about the numerical value for $\Gamma(r/2)$)
- (b) If $X \sim N(0, 4)$, what is the distribution of X^2 ?
- (c) Let $Z_1, Z_2, \dots, Z_n$ have standard normal distributions, $N(0, 1)$. If these random variables are independent, show is the distribution of $W = Z_1^2 + Z_2^2 + \dots + Z_n^2$?
- (d) Using (c), prove that if $Z_1, Z_2, \dots, Z_n$ have distributions, $N(\mu_i, \sigma_i^2)$ and independent, then the distribution of $W = \sum_{i=1}^n \frac{(Z_i - \mu_i)^2}{\sigma_i^2}$ is $\chi^2(n)$.

Hint: you might find useful to review sections 3.3 and 5.4 of the textbook.

Problem 6. Show that for any data samples $X_1, \dots, X_n$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

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Let $\tilde{X} := \sum_{i=1}^n X_i^2$. Represent s^2 in terms of $\bar{X}$ and $\tilde{X}$.

Problem 7. 6.1-2 from Hogg et al textbook.

Problem 8. Using table Va-b from Appendix B of the textbook, explain percentages that appear in the Empirical Rule (68%, 95% and 99.7%, see page 229), in the case when your data is well approximated by the Normal distribution $N(0, 1)$.